

Day 4: Hands-on Subnetting Practical (1 hour): Divide a network into subnets and assign IP addresses to devices. Task: Document the steps to create a subnet plan.

Task 1.

Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

- Network: 192.168.10.0/24
- Required subnets: 4
- , so we borrow 2 bits from the host portion.
- New CIDR: /26
- Subnet mask: 255.255.255.192
- Addresses per subnet:
- Usable hosts per subnet: $64 - 2 = 62$

Subnet	Network Address	Usable IP Range	Broadcast Address
Subnet 1	192.168.10.0/26	192.168.10.1 – 192.168.10.62	192.168.10.63
Subnet 2	192.168.10.64/26	192.168.10.65 – 192.168.10.126	192.168.10.127
Subnet 3	192.168.10.128/26	192.168.10.129 – 192.168.10.190	192.168.10.191
Subnet 4	192.168.10.192/26	192.168.10.193 – 192.168.10.254	192.168.10.255

Task 2.

For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Hint: Make sure not to use the network or broadcast address for any device.

Network: 192.168.10.0/26

Usable range: 192.168.10.1 – 192.168.10.62

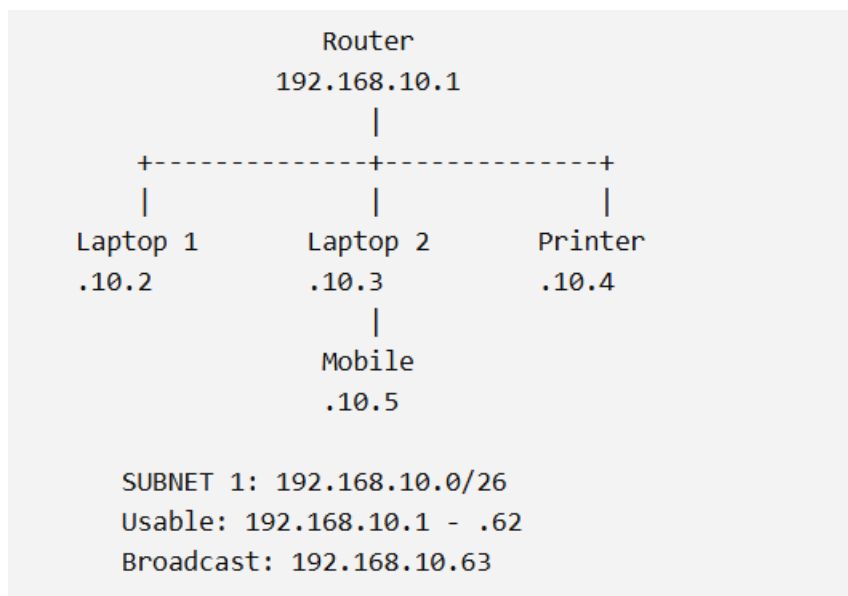
Broadcast: 192.168.10.63

Default Gateway :192.168.10.1

Device	IP Address	Subnet Mask
Router/Gateway	192.168.10.1	255.255.255.192
Laptop 1	192.168.10.2	255.255.255.192
Laptop 2	192.168.10.3	255.255.255.192
Printer	192.168.10.4	255.255.255.192
Mobile	192.168.10.5	255.255.255.192

Task 3.

Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.



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Subnet 2: 192.168.10.64/26
Range:    192.168.10.65 - 192.168.10.126
Broadcast:192.168.10.127

Subnet 3: 192.168.10.128/26
Range:    192.168.10.129 - 192.168.10.190
Broadcast:192.168.10.191

Subnet 4: 192.168.10.192/26
Range:    192.168.10.193 - 192.168.10.254
Broadcast:192.168.10.255
```

Task 4.

Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Step 1: Start with the original network

We have:

192.168.10.0/24

A /24 means:

Network bits = 24

Host bits = 8

So the original network has:

$2^8 = 256$ total addresses

Step 2: Decide how many subnets we need

We need 4 subnets.

Find the number of bits needed:

$2^2 = 4$

Therefore, we need to borrow 2 host bits.

Step 3: Calculate the new subnet prefix

Original prefix:

/24

Borrow 2 bits:

$24 + 2 = /26$

Therefore, our new subnet is:

/26

Step 4: Convert /26 to subnet mask

A /26 mask is:

11111111.11111111.11111111.11000000

Converting to decimal:

255.255.255.192

Therefore:

Subnet mask = 255.255.255.192

Step 5: Calculate hosts per subnet

A /26 leaves:

$32 - 26 = 6$ host bits

Total addresses:

$2^6 = 64$

Two addresses cannot normally be assigned to devices:

- Network address
- Broadcast address

Therefore:

$64 - 2 = 62$ usable hosts

So each subnet supports 62 usable devices.

Step 6: Find the subnet increment

The interesting octet is the last octet.

$$256 - 192 = 64$$

Therefore, our subnet increment is 64.

Start at 0 and keep adding 64:

0

$$0 + 64 = 64$$

$$64 + 64 = 128$$

$$128 + 64 = 192$$

So our four network addresses are:

192.168.10.0

192.168.10.64

192.168.10.128

192.168.10.192

Step 7: Find broadcast addresses

The next subnet starts immediately after the previous broadcast.

Therefore:

Subnet 1:

Network = .0

Broadcast = .63

Subnet 2:

Network = .64

Broadcast = .127

Subnet 3:

Network = .128

Broadcast = .191

Subnet 4:

Network = .192

Broadcast = .255

Step 8: Find usable host ranges

The usable range starts one address after the network address and ends one address before the broadcast address.

For example:

Subnet 1

Network: 192.168.10.0

First host: 192.168.10.1

Last host: 192.168.10.62

Broadcast: 192.168.10.63

Therefore, for our five devices:

Router → 192.168.10.1

Laptop 1 → 192.168.10.2

Laptop 2 → 192.168.10.3

Printer → 192.168.10.4

Mobile → 192.168.10.5